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%% Advanced Battery Project Visualization Suite
% This script generates advanced diagrams for a 72V Battery Pack project.
% Includes: Transient Discharge Profiles, Chemistry Radar Plots, and SOH
Degradation.

clear; clc; close all;

%% ======================================================================
%% DIAGRAM 1: DUAL Y-AXIS DYNAMIC DISCHARGE PROFILE (25A Profile)
%% ======================================================================
figure('Name', 'Transient Discharge Profile', 'Color', 'w', 'Position',
[100, 100, 800, 450]);

% Simulate 1 hour (3600 seconds) of a 25A constant current discharge
time_sec = 0:10:3600;

% Mathematical modeling of an actual 72V pack voltage drop down to ~60V
cutoff
voltage = 72.42 - 10.5 * (time_sec / 3600).^0.3 - 1.8 * (time_sec / 3600).^4;

% Mathematical modeling of thermal rise stabilizing at 42.1 deg C
ambient_temp = 25.0;
max_stabilized_temp = 42.1;
temperature = ambient_temp + (max_stabilized_temp - ambient_temp) * (1 -
exp(-time_sec / 1000));

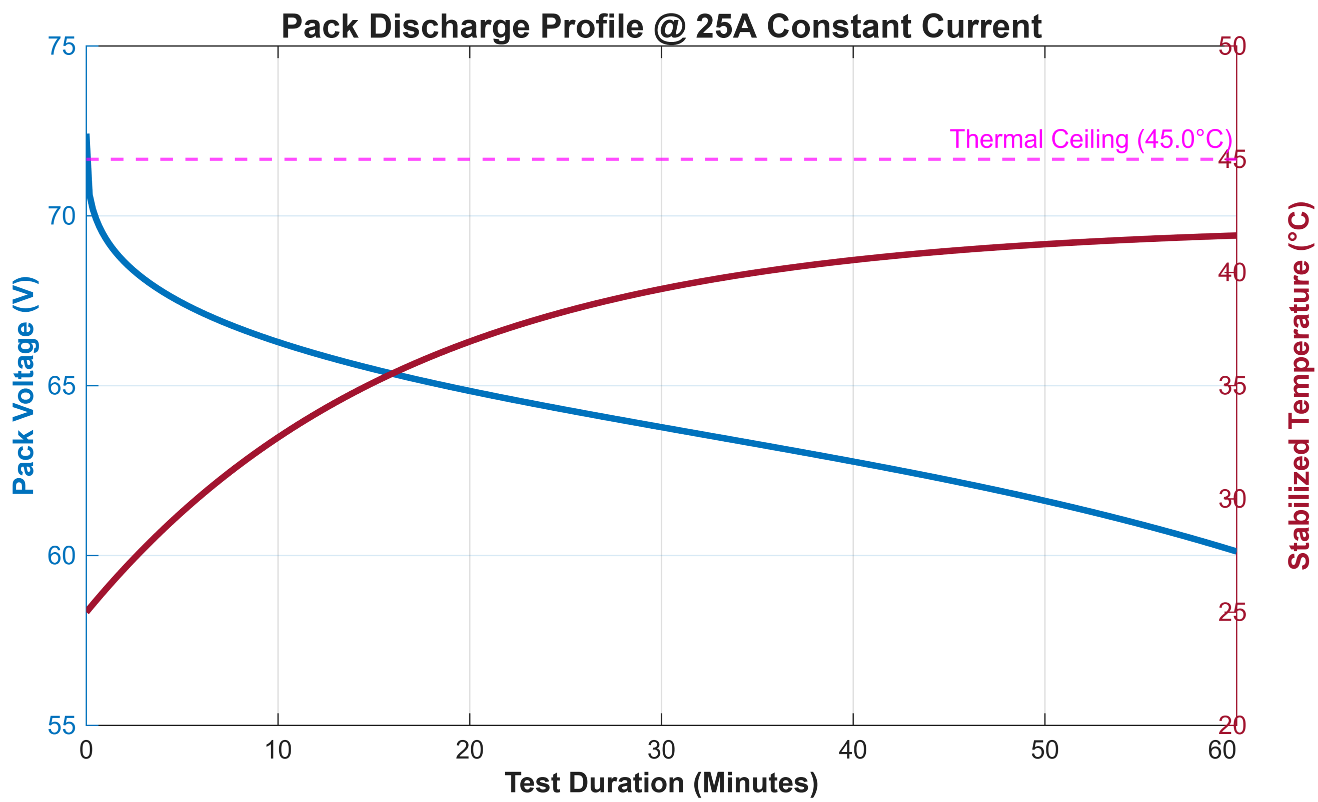
% Plotting Left Axis: Voltage
yyaxis left
plot(time_sec/60, voltage, 'LineWidth', 2.5, 'Color', '#0072BD');
ylabel('Pack Voltage (V)', 'FontWeight', 'bold', 'FontSize', 11);
ylim([55, 75]);
grid on;

% Plotting Right Axis: Temperature
yyaxis right
plot(time_sec/60, temperature, 'LineWidth', 2.5, 'Color', '#A2142F');
ylabel('Stabilized Temperature (\circC)', 'FontWeight', 'bold', 'FontSize',
11);
ylim([20, 50]);

% Critical Operational Threshold Lines
hold on;
yline(60.0, '--r', 'Low UV Trip (60.0V)', 'LabelHorizontalAlignment',
'left', 'LineWidth', 1.2);
yline(45.0, '--m', 'Thermal Ceiling (45.0\circC)',
'LabelHorizontalAlignment', 'right', 'LineWidth', 1.2);

% Formatting
xlabel('Test Duration (Minutes)', 'FontWeight', 'bold', 'FontSize', 11);
```

```
title('Pack Discharge Profile @ 25A Constant Current', 'FontSize', 13,
'FontWeight', 'bold');
ax = gca;
ax.YAxis(1).Color = '#0072BD';
ax.YAxis(2).Color = '#A2142F';
```

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%% ========================================================================
%% DIAGRAM 2: MULTI-ATTRIBUTE RADAR (SPIDER) CHART
%% ========================================================================
figure('Name', 'Chemistry Trade-Off Matrix', 'Color', 'w', 'Position', [150,
150, 600, 500]);

% Define 5 core battery attributes (Normalized Rating Scale: 1 to 5)
% Metrics order: [Specific Energy, Safety, Cycle Life, C-Rate Peak, Cost
Efficiency]
nmc_metrics = [5, 2, 2, 5, 3];
lfp_metrics = [2, 5, 5, 3, 5];

% Append the first value to the end to close the radar polygon loop
nmc_plot = [nmc_metrics, nmc_metrics(1)];
lfp_plot = [lfp_metrics, lfp_metrics(1)];

% Calculate equiangular spacing for a 5-sided polygon
theta = linspace(0, 2*pi, 6);
```

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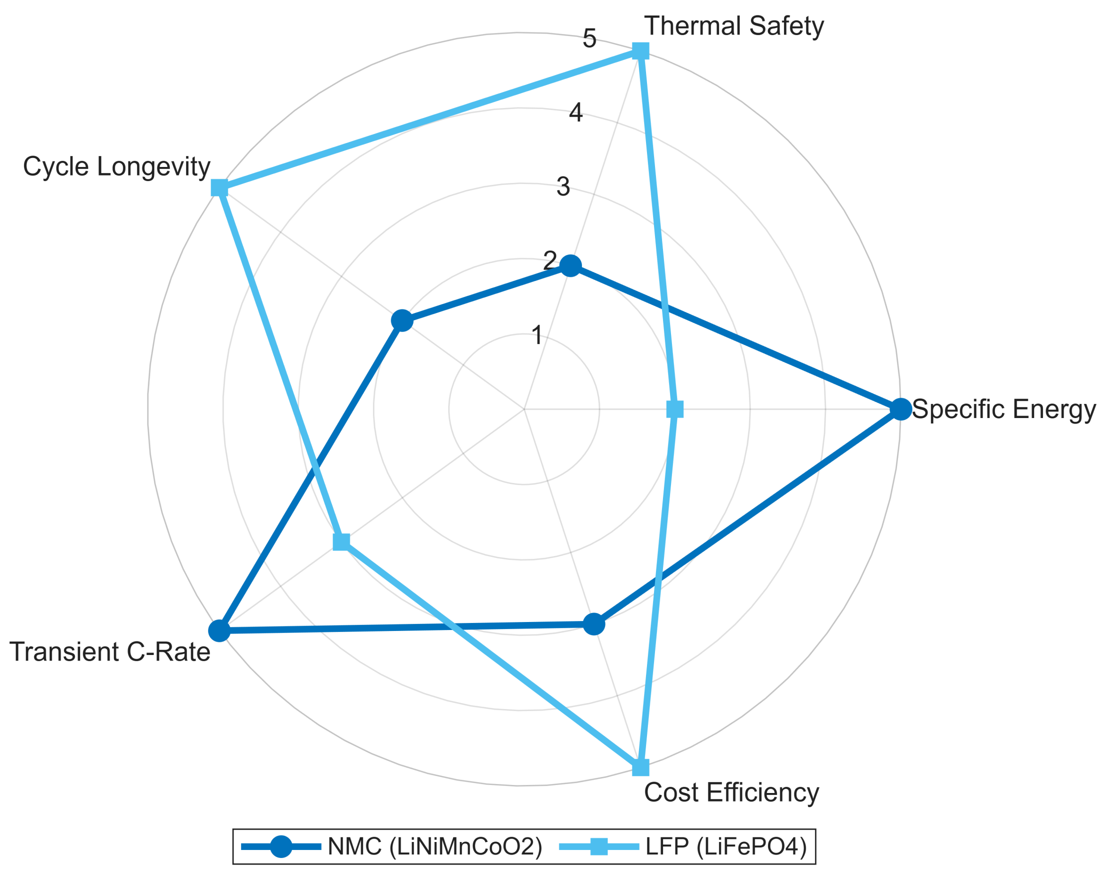
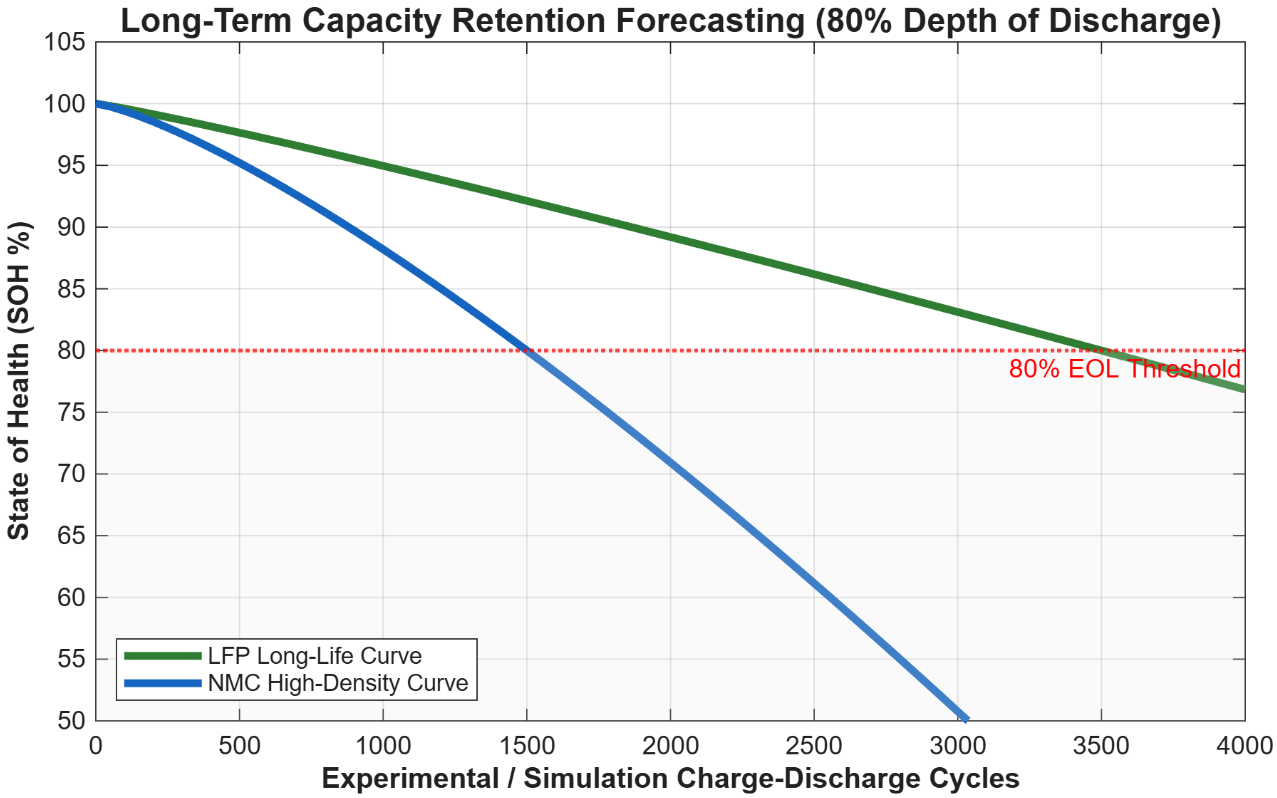
% Create polar plot
polarplot(theta, nmc_plot, '-o', 'LineWidth', 2.5, 'Color', '#0072BD',
'MarkerFaceColor', '#0072BD');
hold on;
polarplot(theta, lfp_plot, '-s', 'LineWidth', 2.5, 'Color', '#4DBEEF',
'MarkerFaceColor', '#4DBEEF');

% Diagram styling and labeling
r_labels = {'Specific Energy', 'Thermal Safety', 'Cycle Longevity',
'Transient C-Rate', 'Cost Efficiency', ''};
ax = gca;
ax.ThetaTick = rad2deg(theta(1:5));
ax.ThetaTickLabel = r_labels(1:5);
ax.RGrid = true;
ax.RTick = 1:5;
ax.RLim = [0 5];

title('Battery Chemistry Trade-Off Matrix (NMC vs LFP)', 'FontSize', 13,
'FontWeight', 'bold', 'Units', 'normalized', 'Position', [0.5, 1.1, 0]);
legend({'NMC (LiNiMnCoO2)', 'LFP (LiFePO4)'}, 'Location', 'southoutside',
'Orientation', 'horizontal');
```



```matlab
%% ==========================================================================
%% DIAGRAM 3: LONG-TERM CAPACITY DEGRADATION (SOH Fade Curve)
%% ==========================================================================
figure('Name', 'State-of-Health Degradation Projection', 'Color', 'w',
'Position', [200, 200, 800, 450]);

cycles = 0:50:4000;

% Empirical degradation paths mapping out to your design specs
soh_nmc = 100 - 20 * (cycles / 1500).^1.3;
soh_lfp = 100 - 20 * (cycles / 3500).^1.1;

% Clip negative/unrealistic values below 40% for plotting cleanliness
soh_nmc(soh_nmc < 40) = NaN;

% Plot degradation curves
plot(cycles, soh_lfp, 'LineWidth', 3, 'Color', '#2E7D32'); % Deep Green for
Eco/LFP
hold on;
plot(cycles, soh_nmc, 'LineWidth', 3, 'Color', '#1565C0'); % Deep Blue for
NMC

% 80% End-of-Life (EOL) Boundary Line
yline(80, ':r', '80% EOL Threshold', 'LineWidth', 1.5,
'LabelVerticalAlignment', 'bottom');

% Shade the targeted operational validation zone
fill([0 4000 4000 0], [100 100 800 800]/10, [0.9 0.9 0.9], 'FaceAlpha', 0.2,
'LineStyle', 'none');

% Plot formatting
grid on;
xlabel('Experimental / Simulation Charge-Discharge Cycles', 'FontWeight',
'bold', 'FontSize', 11);
ylabel('State of Health (SOH %)', 'FontWeight', 'bold', 'FontSize', 11);
title('Long-Term Capacity Retention Forecasting (80% Depth of Discharge)',
'FontSize', 13, 'FontWeight', 'bold');
xlim([0, 4000]);
ylim([50, 105]);
legend({'LFP Long-Life Curve', 'NMC High-Density Curve'}, 'Location',
'southwest');
```


Long-Term Capacity Retention Forecasting (80% Depth of Discharge)
State of Health (SOH %)
105
100
95
90
85
80
75
70
65
60
55
50
0
500
1000
1500
2000
2500
3000
3500
4000
Experimental / Simulation Charge-Discharge Cycles
80% EOL Threshold
LFP Long-Life Curve
NMC High-Density Curve